Compliance sheet — star-kepler

Everything a supplier qualification, a legal review or an export check usually asks about star-kepler 0.1.1, in one document.

1. Identity

Document	STAR-PKG-STAR-KEPLER-EN · v1.2 · 2026-10-15
Package	star-kepler
Version	0.1.1
Capability	star.orbit.kepler
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_kepler

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE and NOTICE files, keep the copyright notices, and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_kepler-
0.1.1-py3-none-any.whl
```

FILE	SHA-256
star_kepler-0.1.1-py3-none-any.whl	bf6dc03cb3df336cd833b18861318cb2833fe01b58c4920ae8ee5138b6f0ee7f
star_kepler.py	e28bacf7f0d3e09a31d6e01fc45472f4421c78bf76f9e30c8587d72a504cb7fa
star_kepler-0.1.1-source.zip	8294299fc667dd00b5a6365f367aa6c3be19c73f244faeea2e3f626278ee350a

5. What it does, and its limits

Kepler propagation with universal variables and Stumpff functions: no orbit-geometry singularity (circular, equatorial, elliptic, hyperbolic), whole periods removed, bracketed root with bounded iterations, explicit refusals.

Two-body motion only (no perturbations). Parabolic orbits use the same series but are not separately validated. Rectilinear orbits are refused. No flight heritage.

- `propagate` returns position and velocity for elliptic and hyperbolic orbits, any consistent units; it reproduces Vallado Example 2-4 within the printed precision (1e-3 km, 1e-5 km/s).
- No orbit-geometry singularity: circular and equatorial orbits are propagated like any other; whole periods are removed on closed orbits, so a thousand revolutions return the initial state to 1e-4 km.
- Accuracy: on 450 known orbits (eccentricity 0 to 0.95, 0 and 10 revolutions) the error is below 1e-6 km (1 mm) and 1e-8 km/s; energy and angular momentum are conserved to 1e-9 relative.
- Failure behaviour: non-finite input, zero position, `mu ⩽ 0`, wrong vector length and rectilinear motion raise `ValueError` before any iteration; an exhausted iteration budget raises `RuntimeError`; the call cannot loop forever (bracketed root, bounded bisection).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before

using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_kepler_0_1_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-kepler}: Two-body propagation with universal variables},
  version = {0.1.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_kepler/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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Related: space activities and export control · acceptable use · evidence and limits · licence texts